

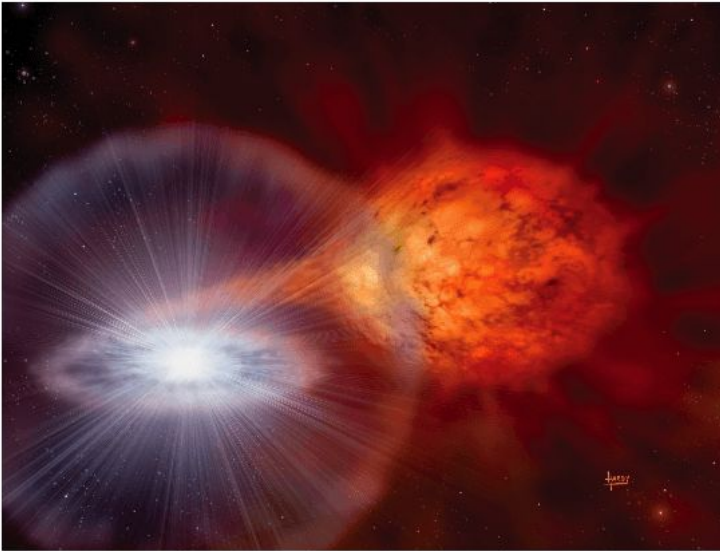
Unveiling the Intrinsic Type Ia Supernova Mass Step

Matt Grayling



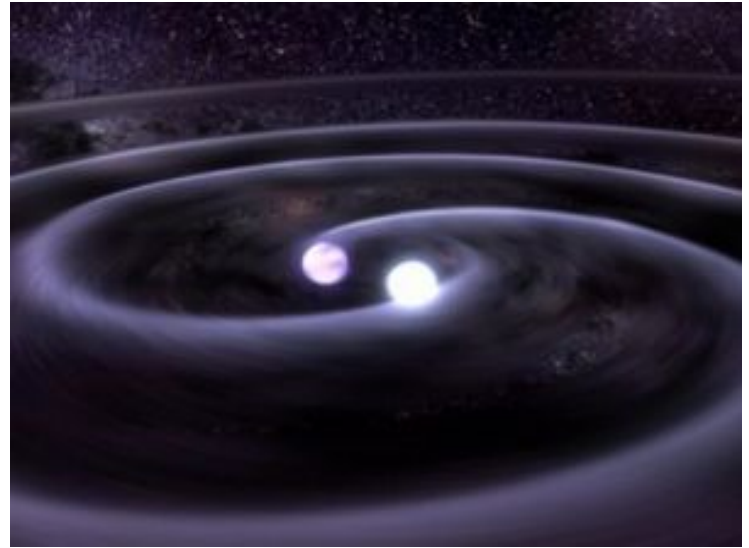
Type Ia Supernovae

Single Degenerate



Credit: David A. Hardy, © David A. Hardy/astroasrt.org

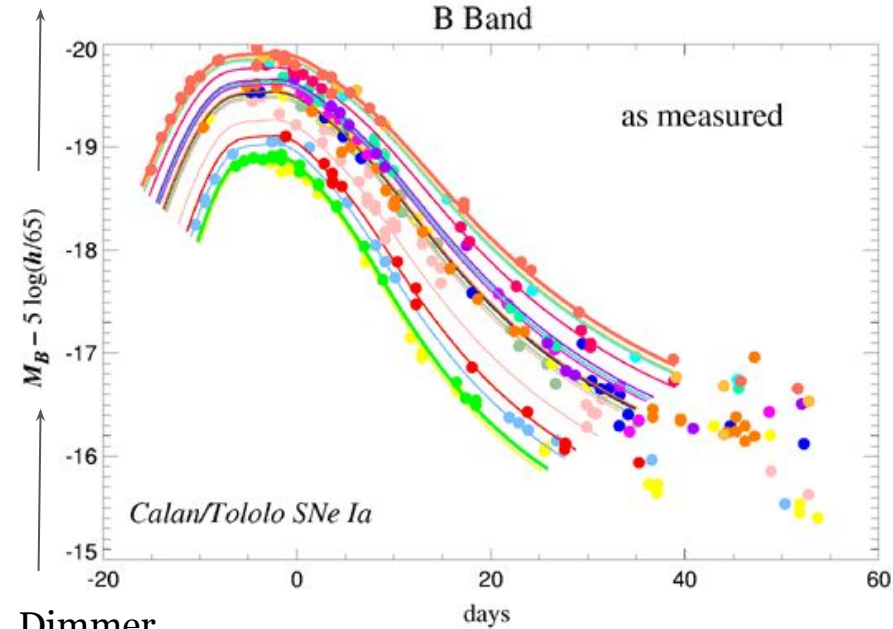
Double Degenerate



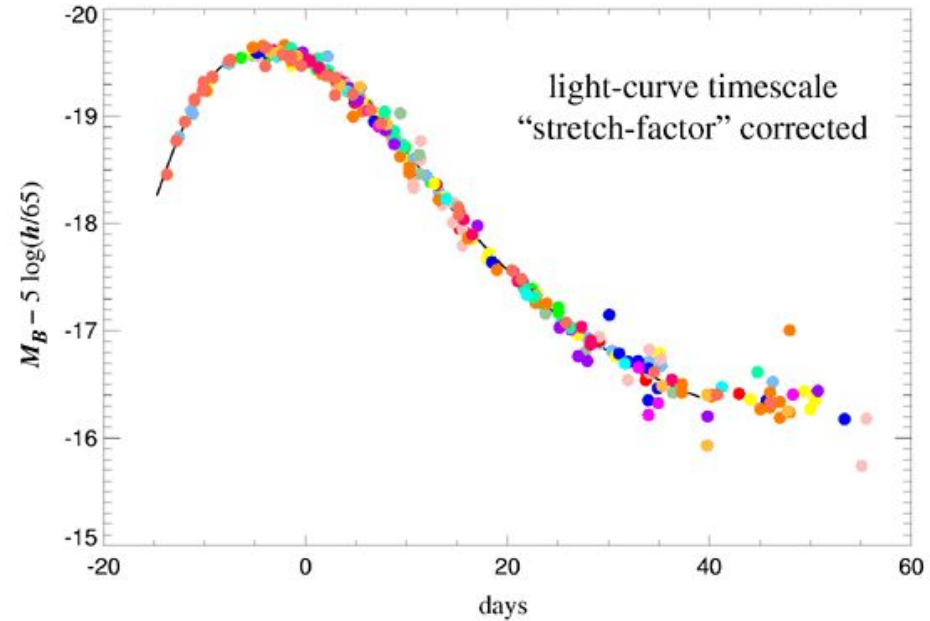
Copyright: GSFC/D. Berry.

Standard(isable) Candles

Brighter



Dimmer

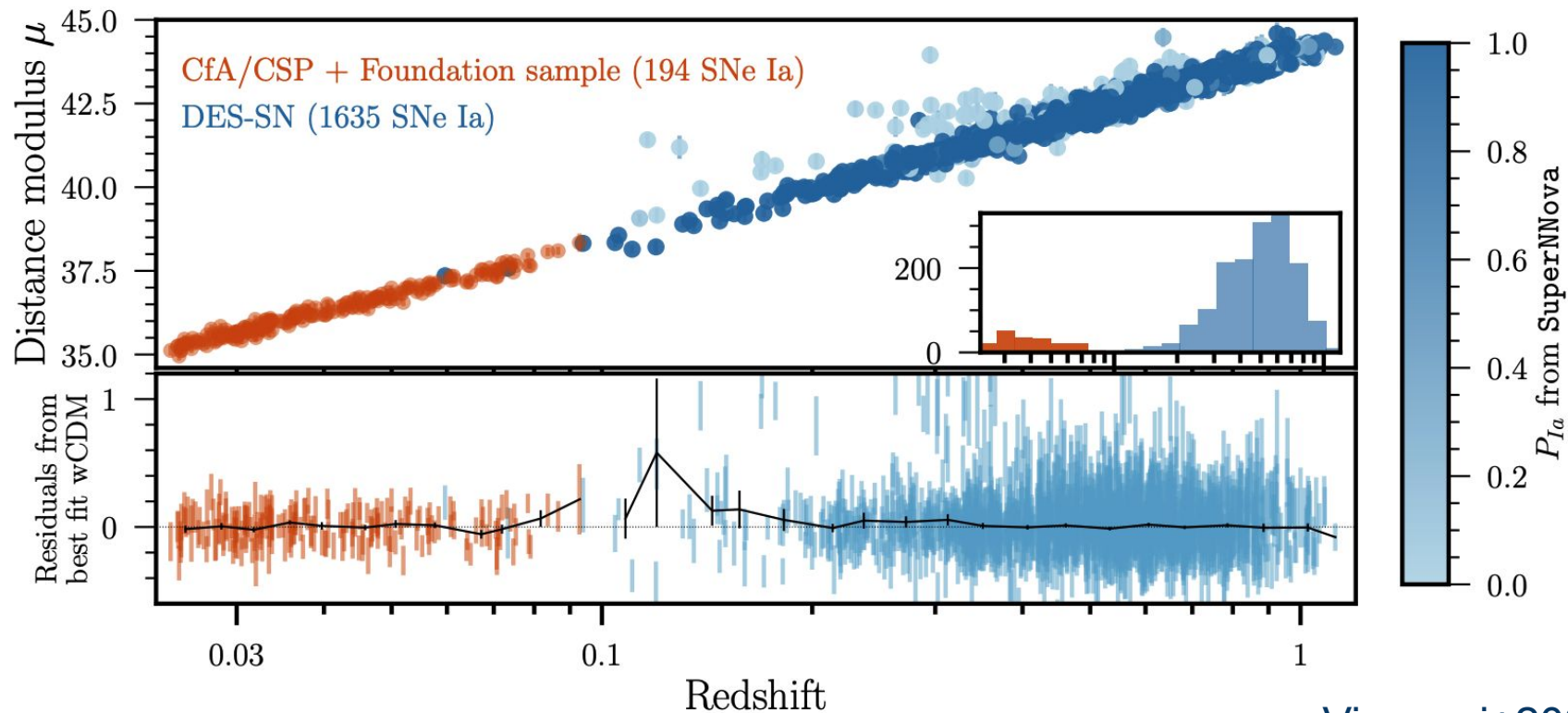


From Supernova Cosmology Project

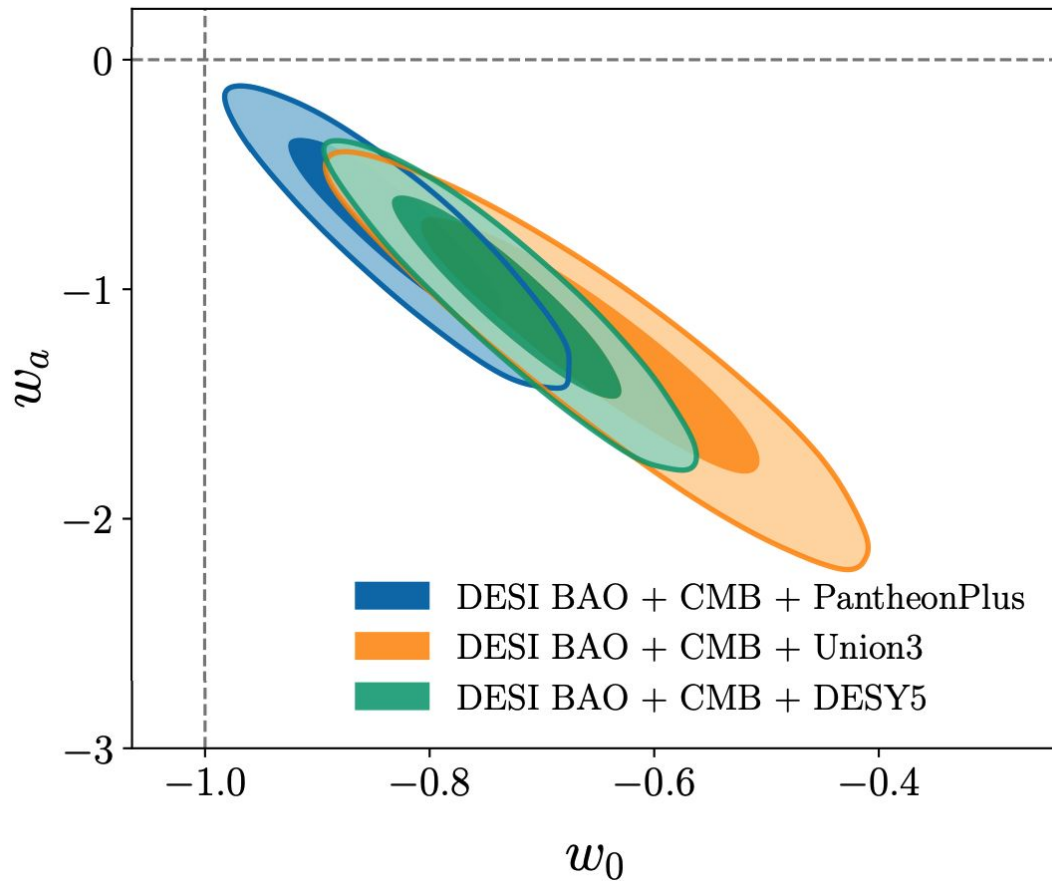
Calculating Distance

$$\mu^s = m_B^s - M + \alpha x_1^s - \beta c^s$$

The Hubble Diagram



Crisis in Cosmology?



DESI Collaboration
2024

Supernova cosmology is easy, right?

The Impact of Dust

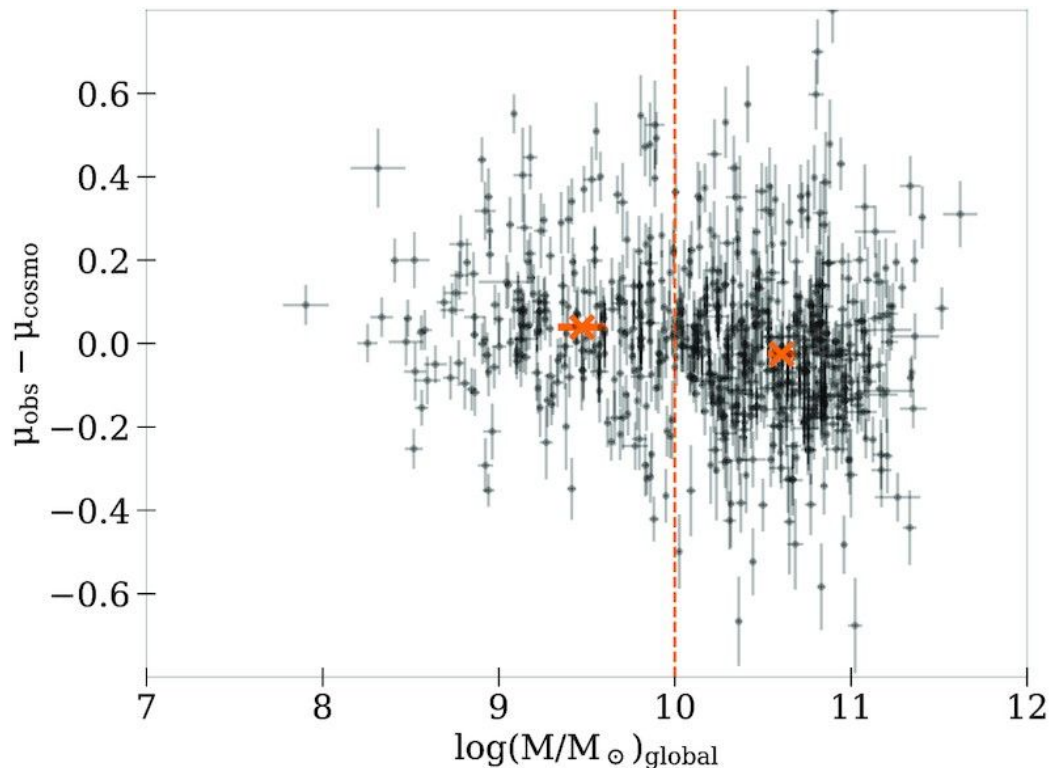
$$\mu^s = m_B^s - M + \alpha x_1^s - \beta c^s$$

$$c^s = c_{\text{int}}^s + E(B - V)^s$$

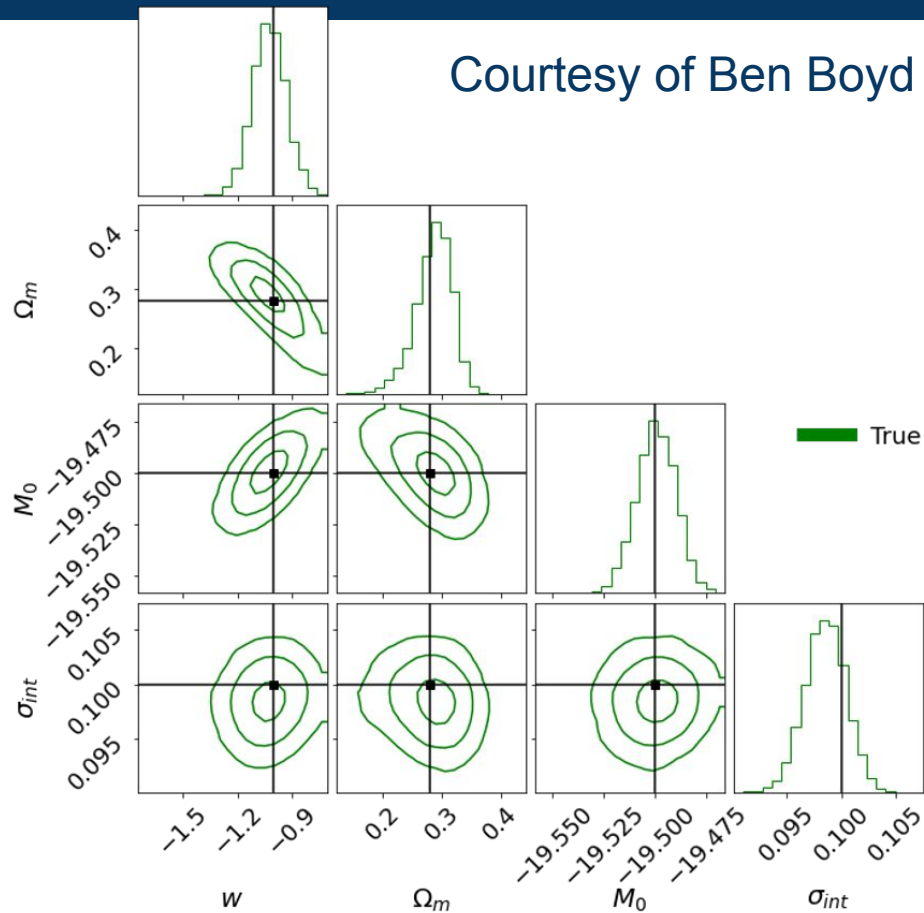
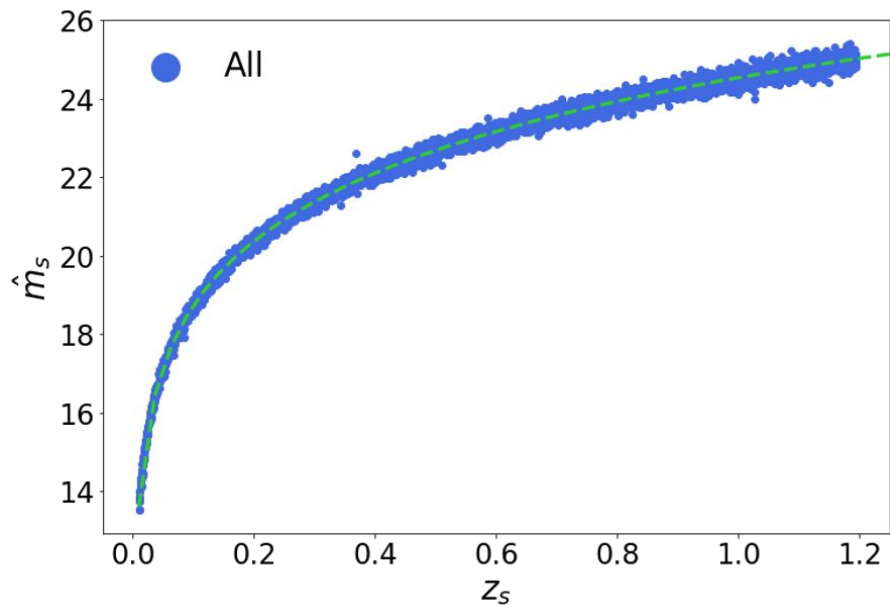

- Colour - one parameter for two separate effects
- Correctly handling dust is key to uncover intrinsic SN properties and accurately estimate distances

The Mass Step

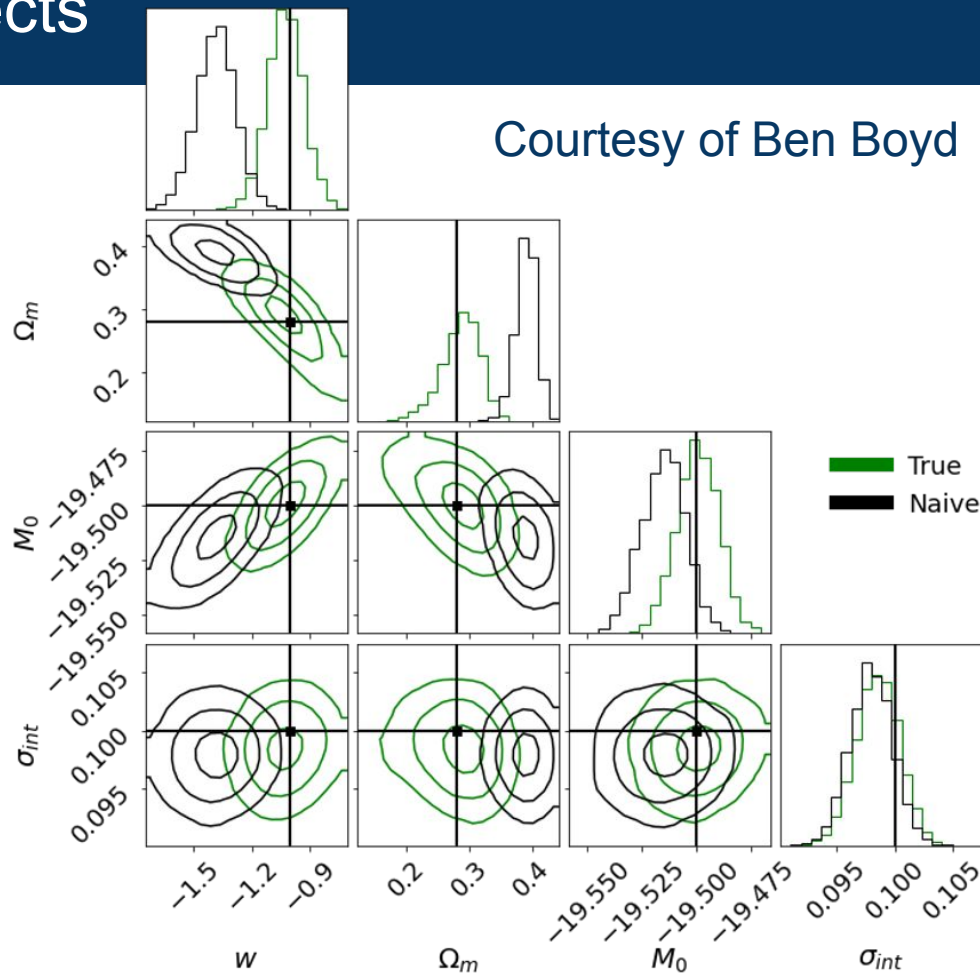
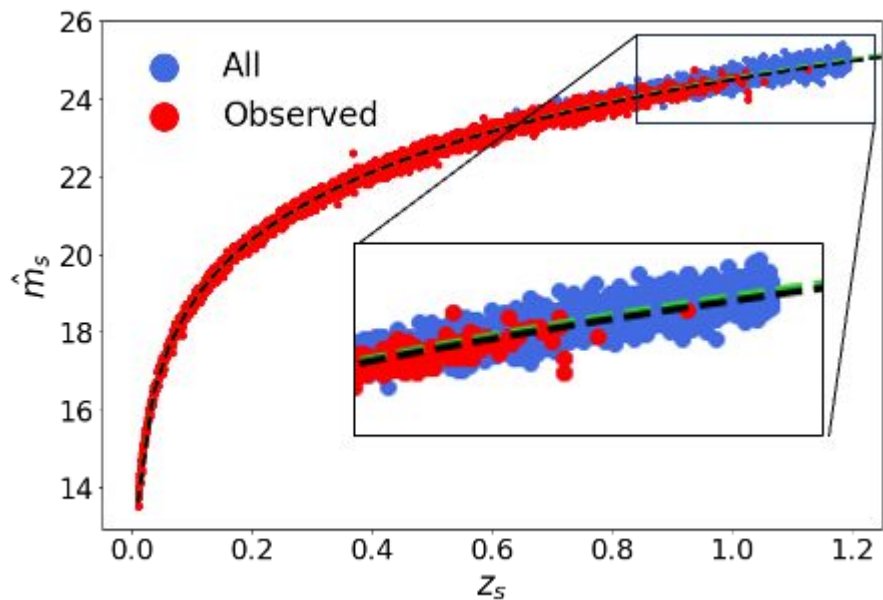
- SN Ia properties depend on environment
- Just different dust properties?
 - SN populations identical, just appear different?
- Or intrinsic differences?
 - Interesting physical implications, different progenitor channels?



The Impact of Selection Effects

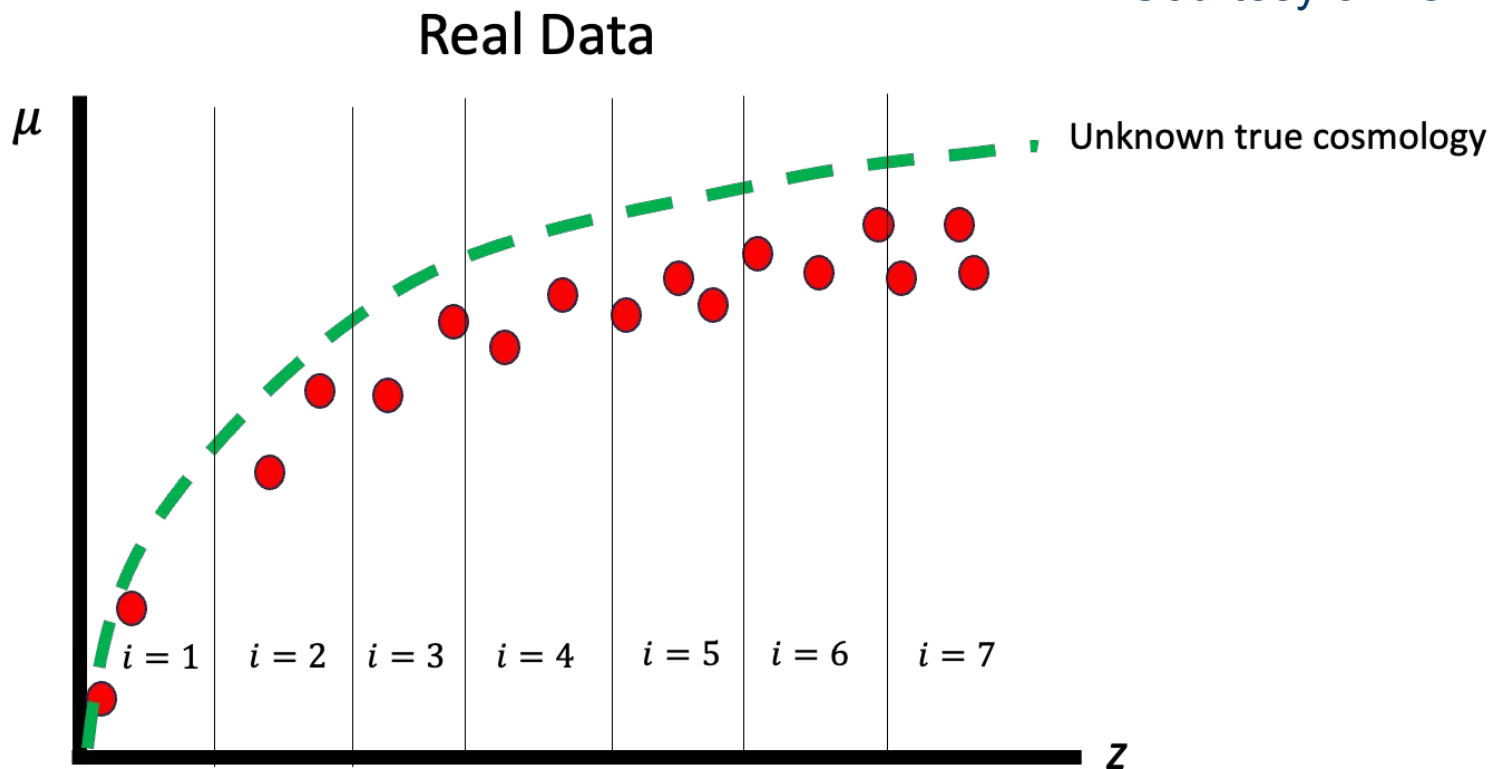


The Impact of Selection Effects



The Impact of Selection Effects

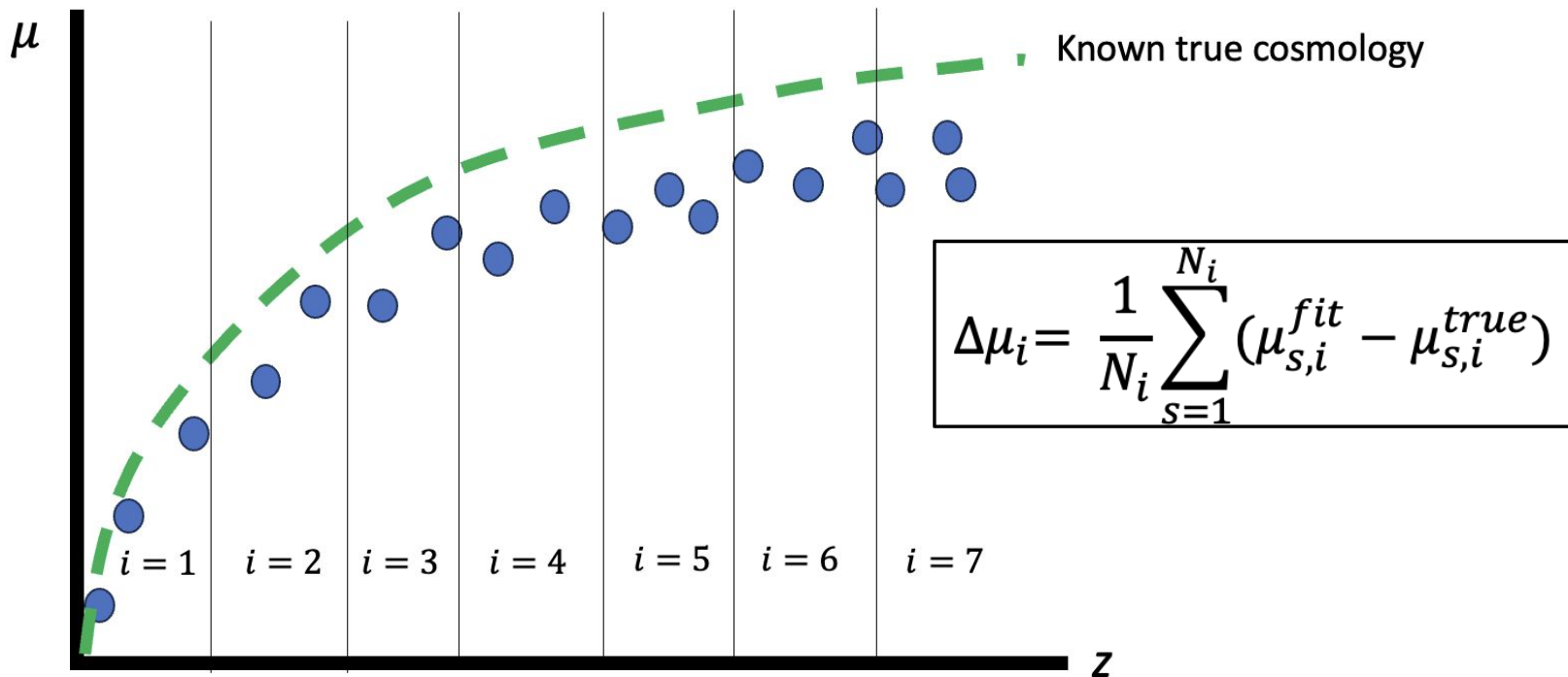
Courtesy of Ben Boyd



The Impact of Selection Effects

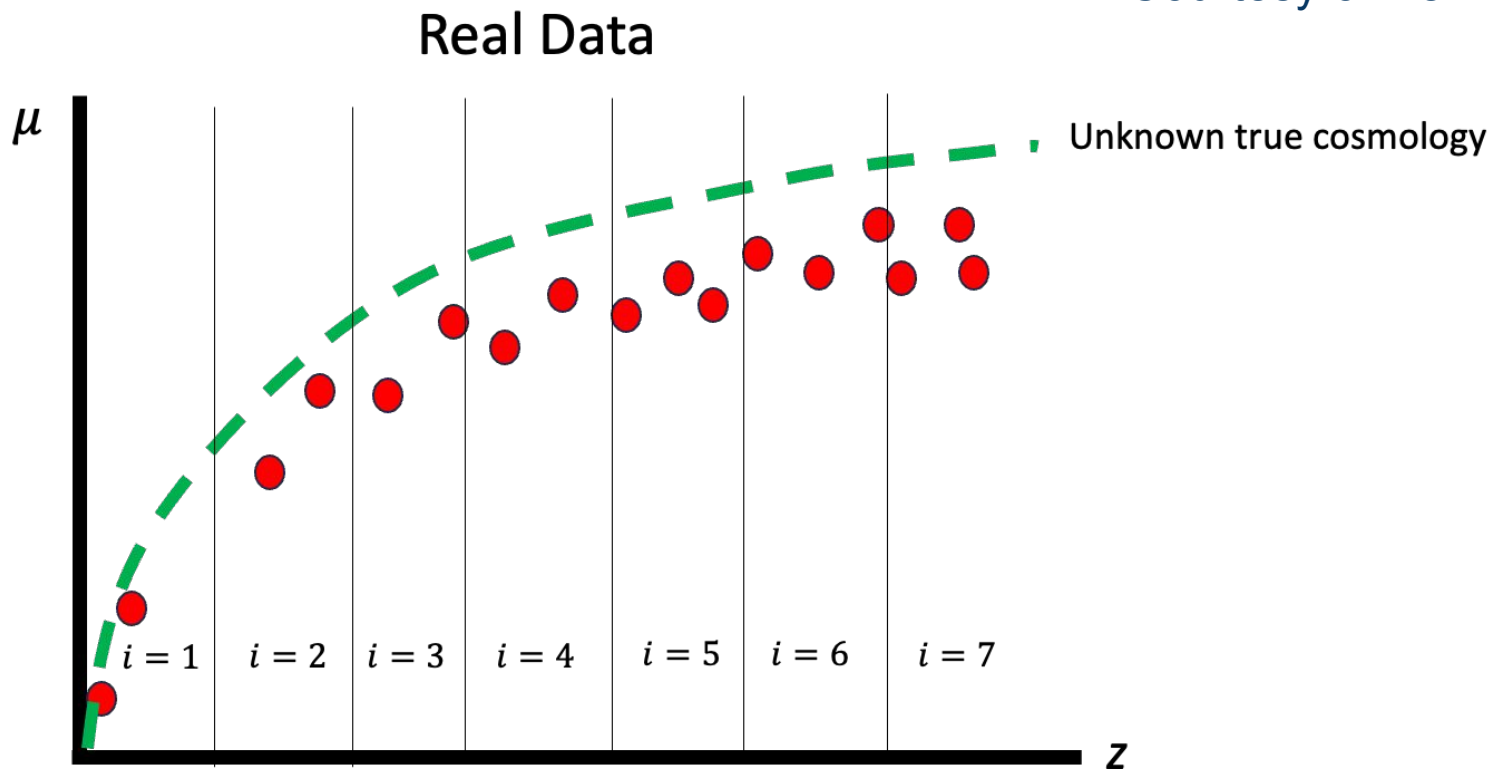
Courtesy of Ben Boyd

SNANA Simulations



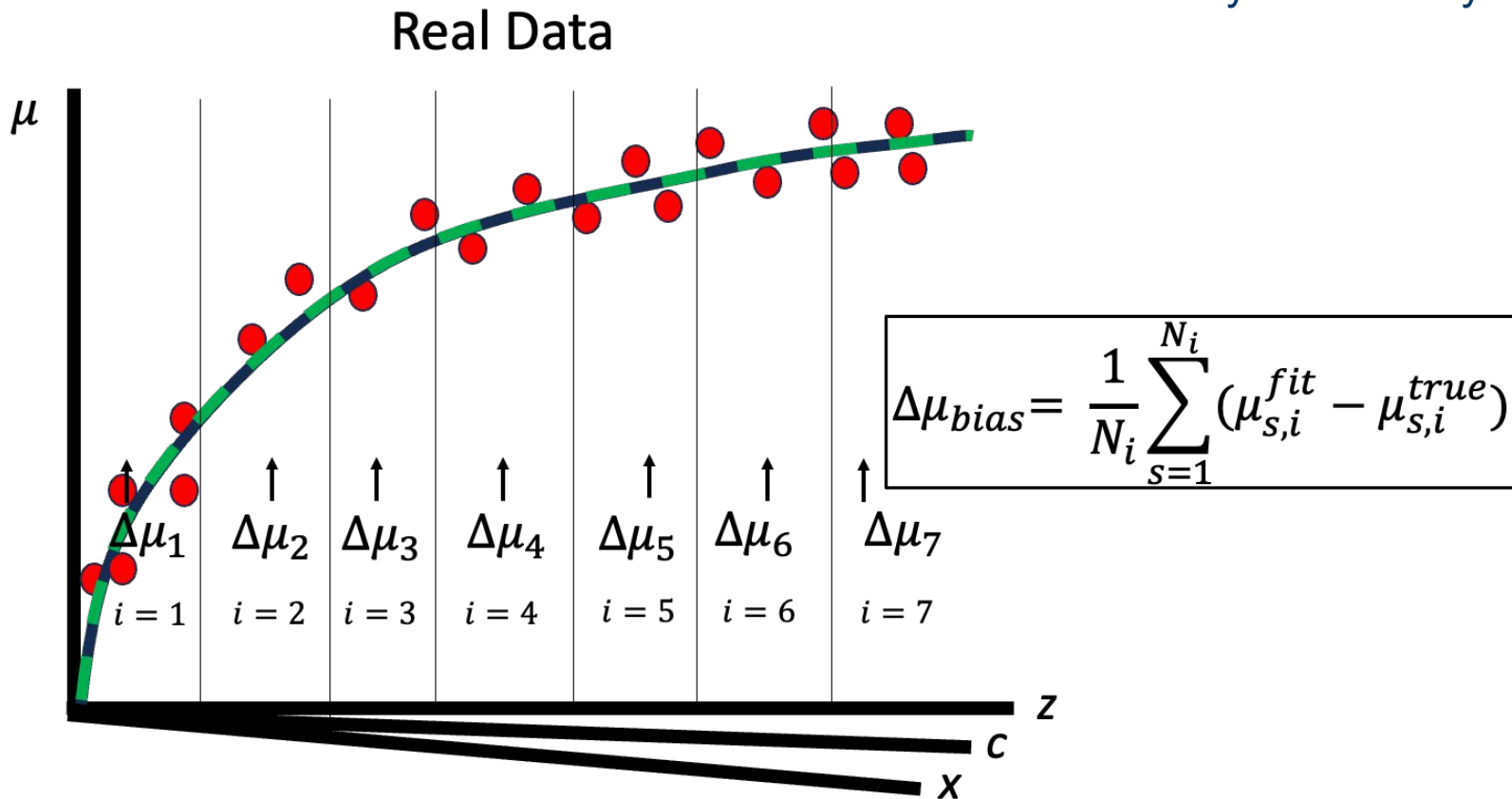
The Impact of Selection Effects

Courtesy of Ben Boyd



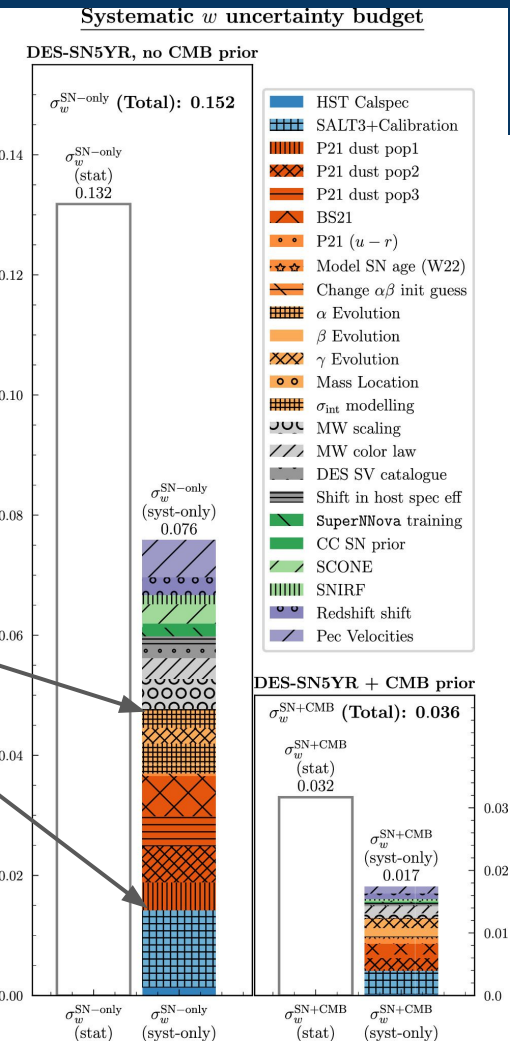
The Impact of Selection Effects

Courtesy of Ben Boyd



Challenges in Supernova Cosmology

The majority of systematic uncertainty in DES5YR relates to unknown astrophysics



Supernova cosmology is inseparably
linked to supernova astrophysics

Recipe for a Type Ia Supernova Model

Type Ia SN SED

=

+

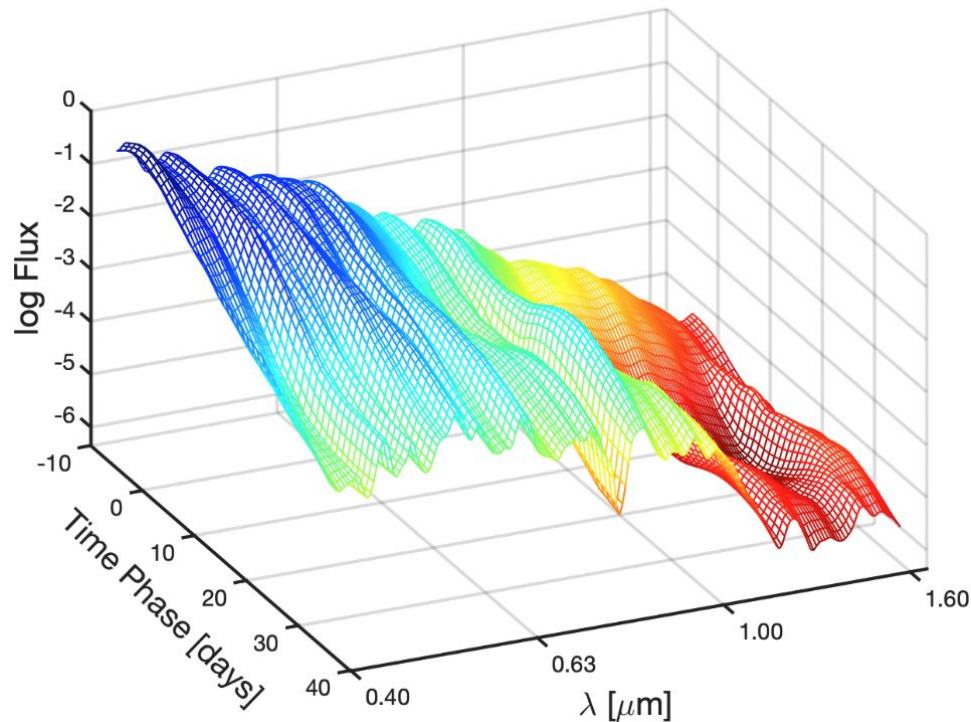
+

+



Recipe for a Type Ia Supernova Model

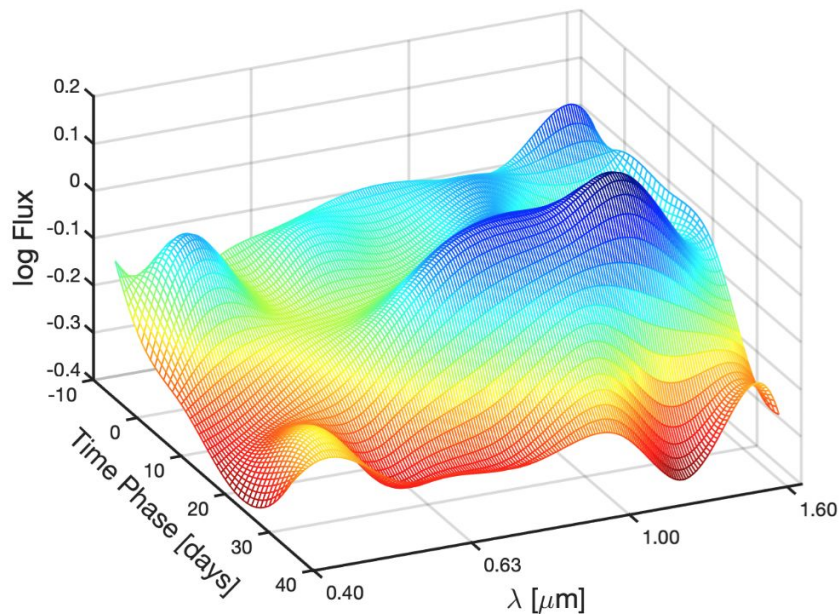
Type Ia SN SED
=
Mean Intrinsic SED
+
+
+



Recipe for a Type Ia Supernova Model

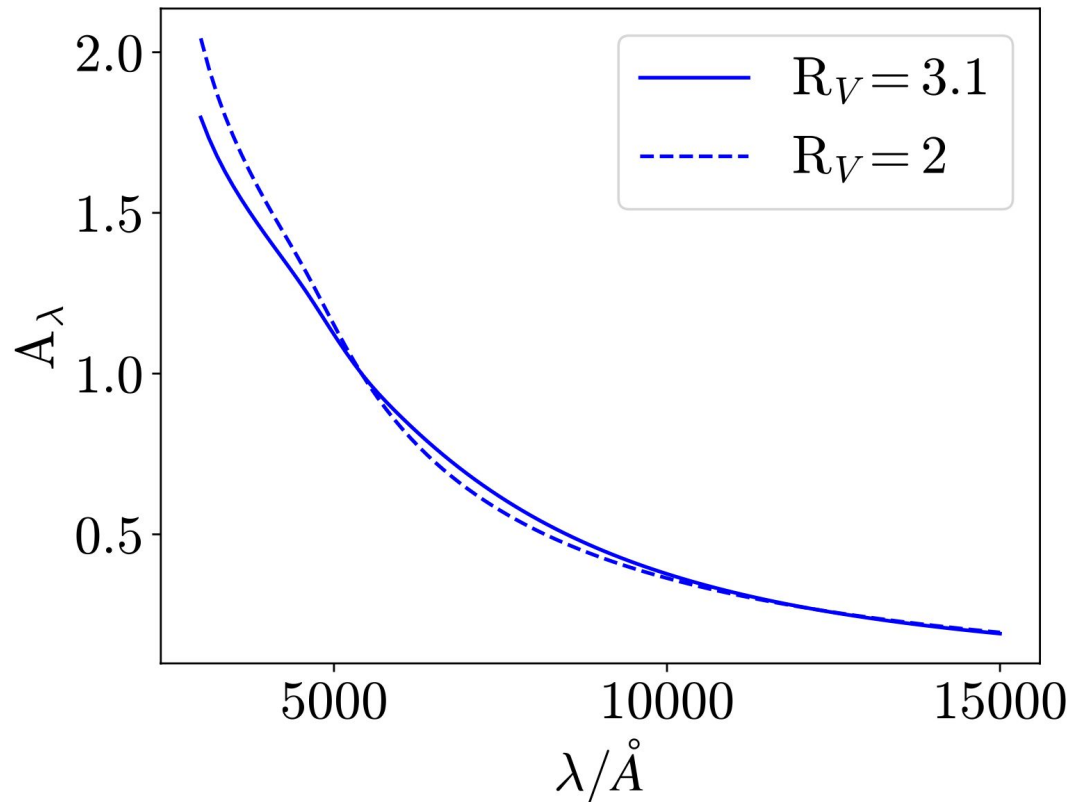
Type Ia SN SED
=
Mean Intrinsic SED
+
Effect of Stretch
+
+

$\theta_1 \times$



Recipe for a Type Ia Supernova Model

Type Ia SN SED
=
Mean Intrinsic SED
+
Effect of Stretch
+
Dust Extinction
+



Recipe for a Type Ia Supernova Model

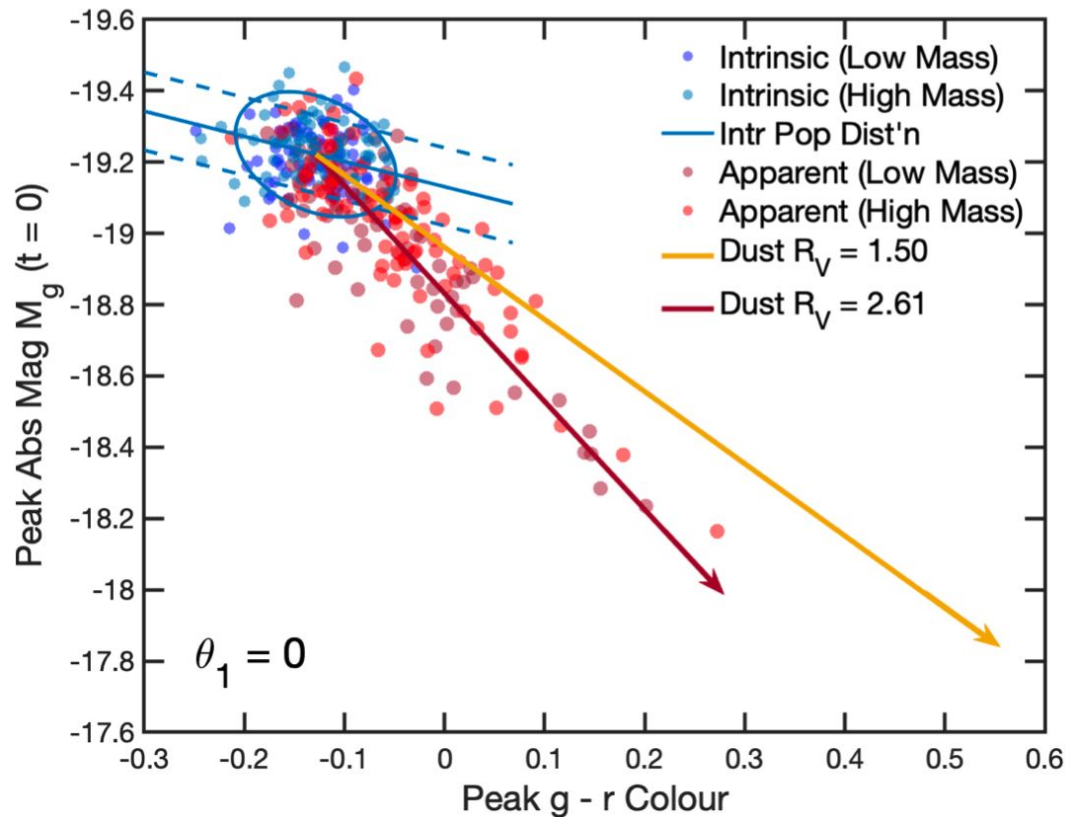
Type Ia SN SED
=
Mean Intrinsic SED
+
Effect of Stretch
+
Dust Extinction
+

$$R_V \sim N(\mu_R, \sigma_R^2)$$

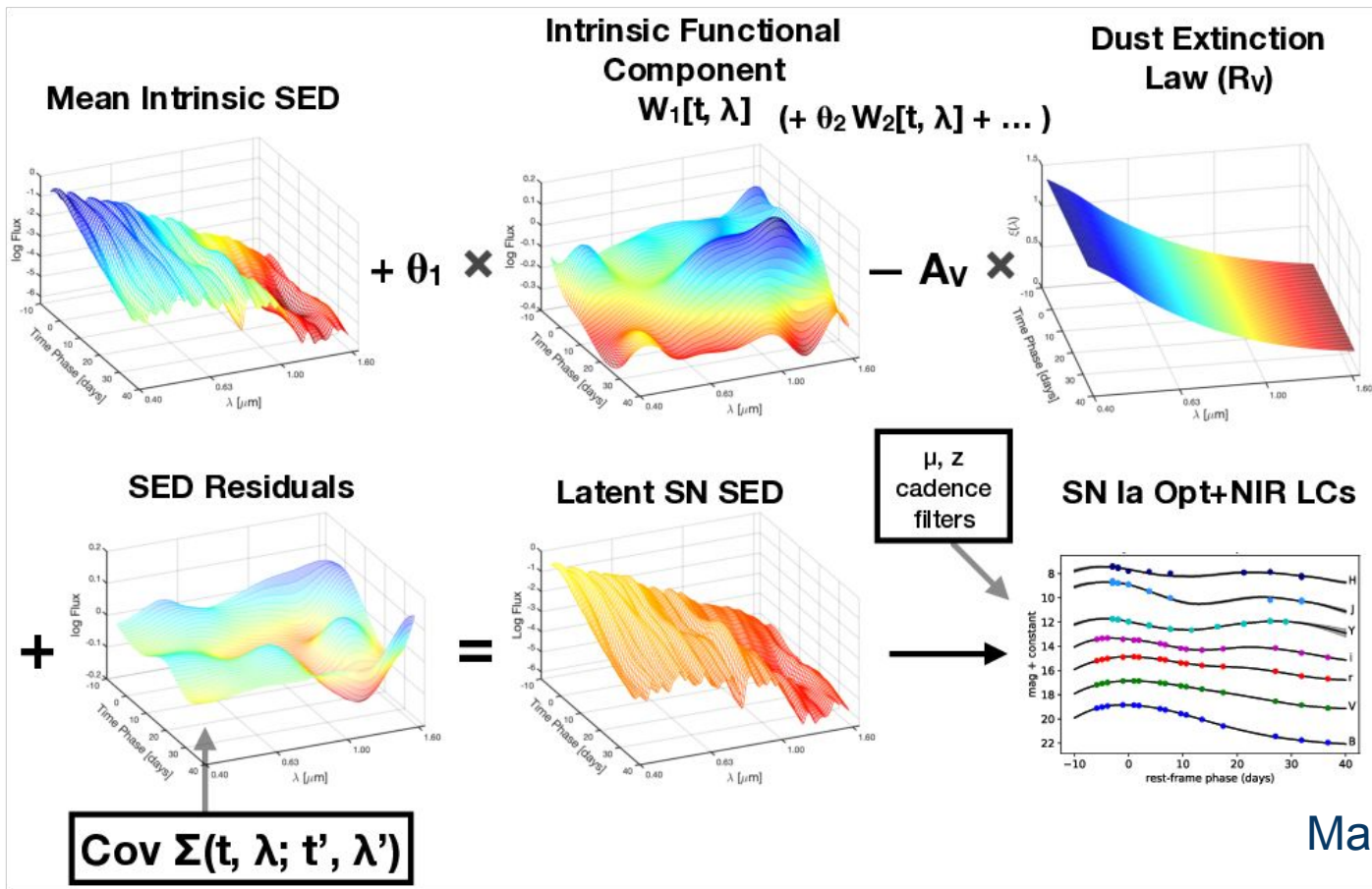
$$A_V \sim \text{Exp}(\tau_A)$$

Recipe for a Type Ia Supernova Model

Type Ia SN SED
=
Mean Intrinsic SED
+
Effect of Stretch
+
Dust Extinction
+
Intrinsic Colour

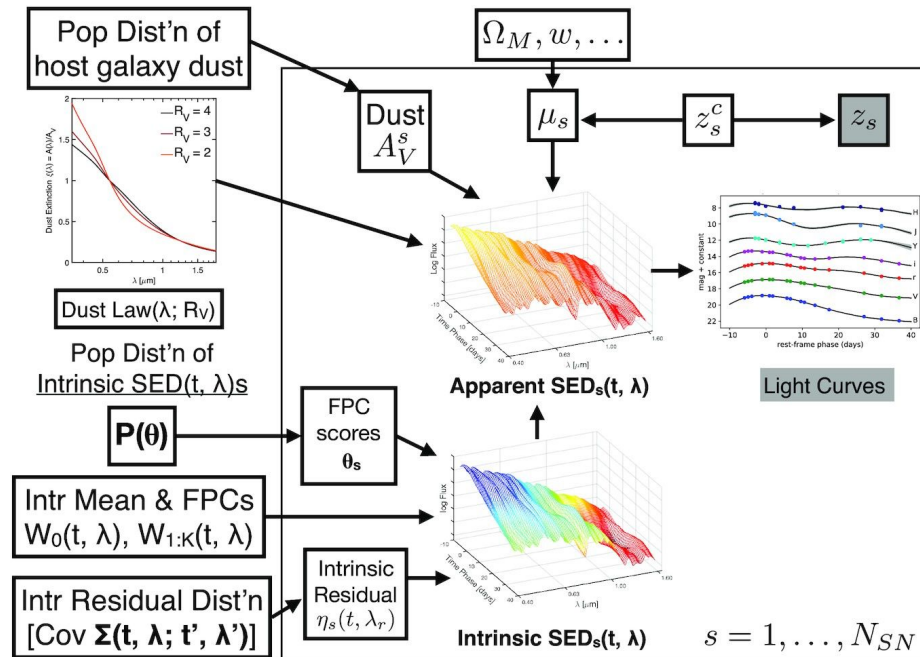


The BayeSN Model



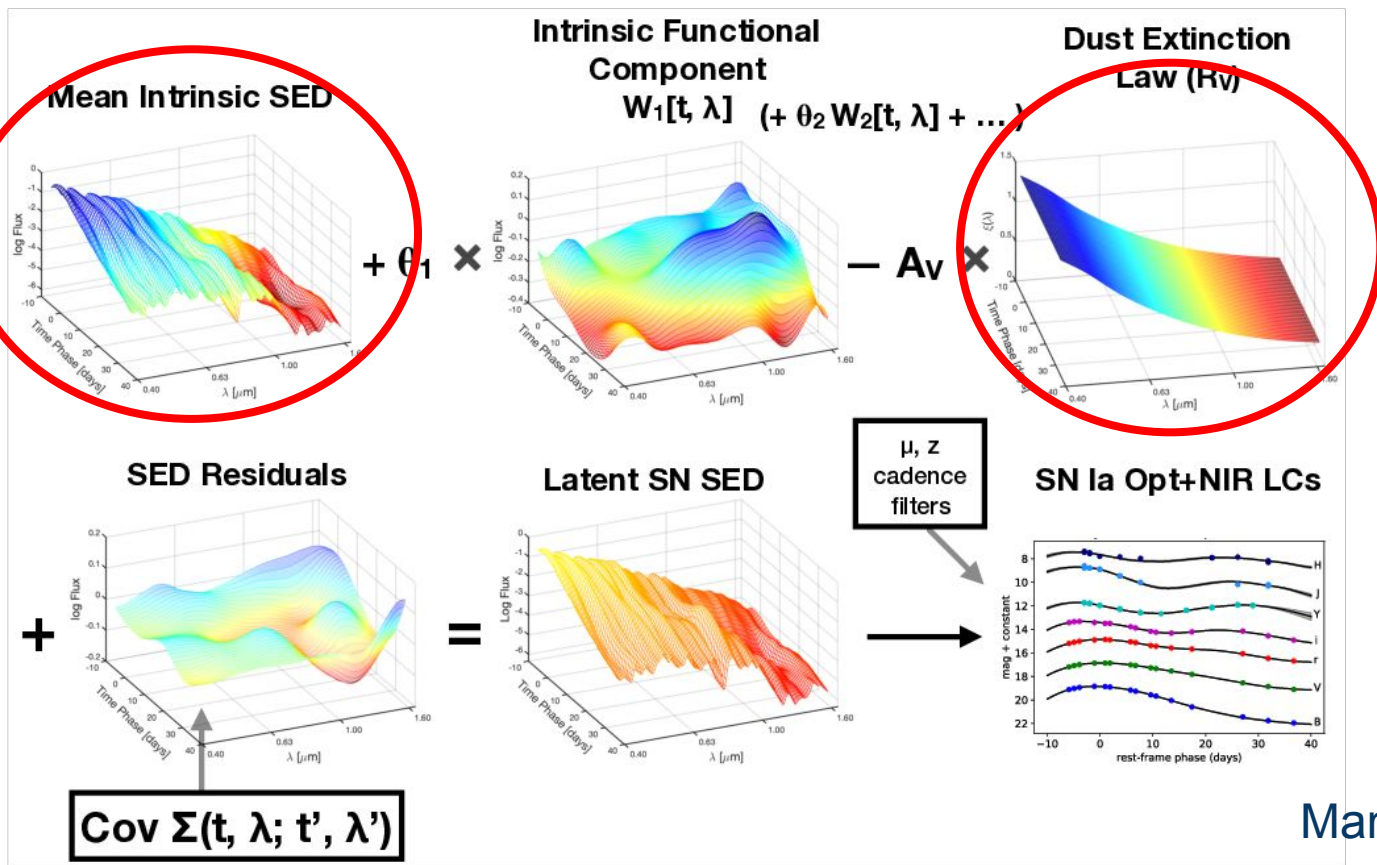
Advantages of BayeSN

- Hierarchical Bayesian Model
 - Allows for separate treatment of dust and intrinsic variation
- Full SED model
 - Uses full light curve in all bands rather than just single colour/magnitude pairs
- Fit for distance jointly with stretch/dust
 - Distance marginalised over dust/intrinsic variation



Mandel+2020

Investigating the Mass Step

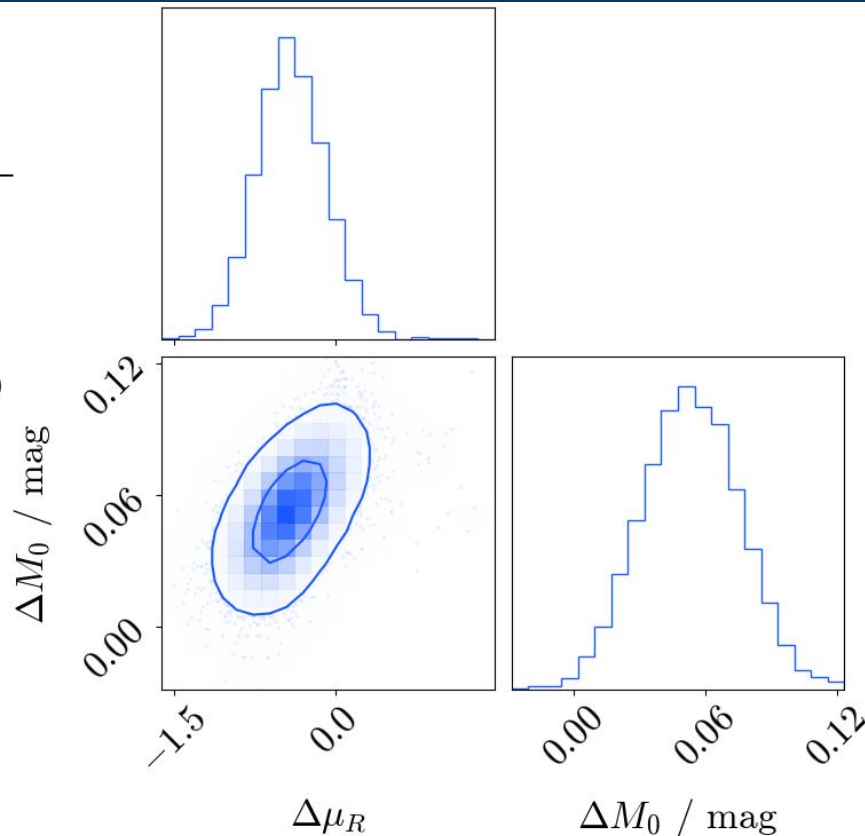


Cause of the Mass Step

Sample	Intrinsic Mass Step / mag
Combined*	0.049 ± 0.019
DES5YR	0.053 ± 0.022

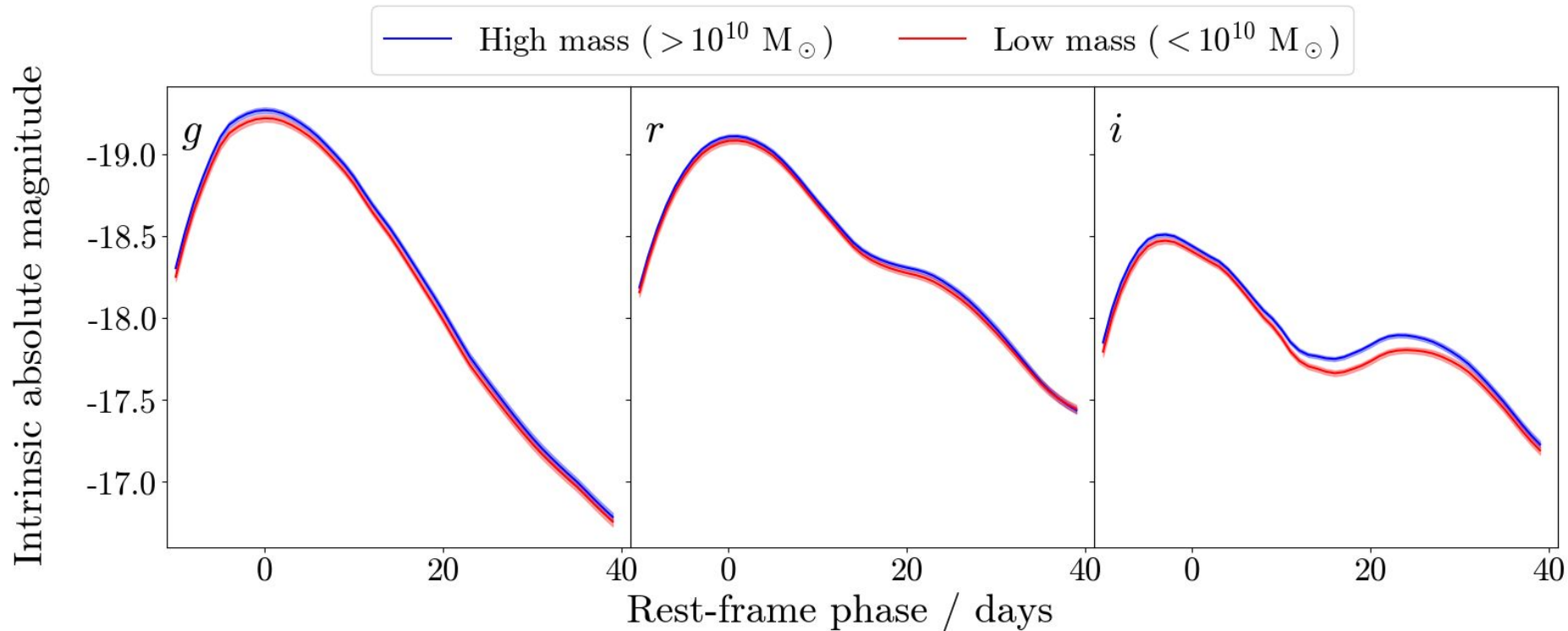
*Combined = DES3YR+Foundation+PS1MD

Dust may play a role, but BayeSN supports an intrinsic contribution to the mass step

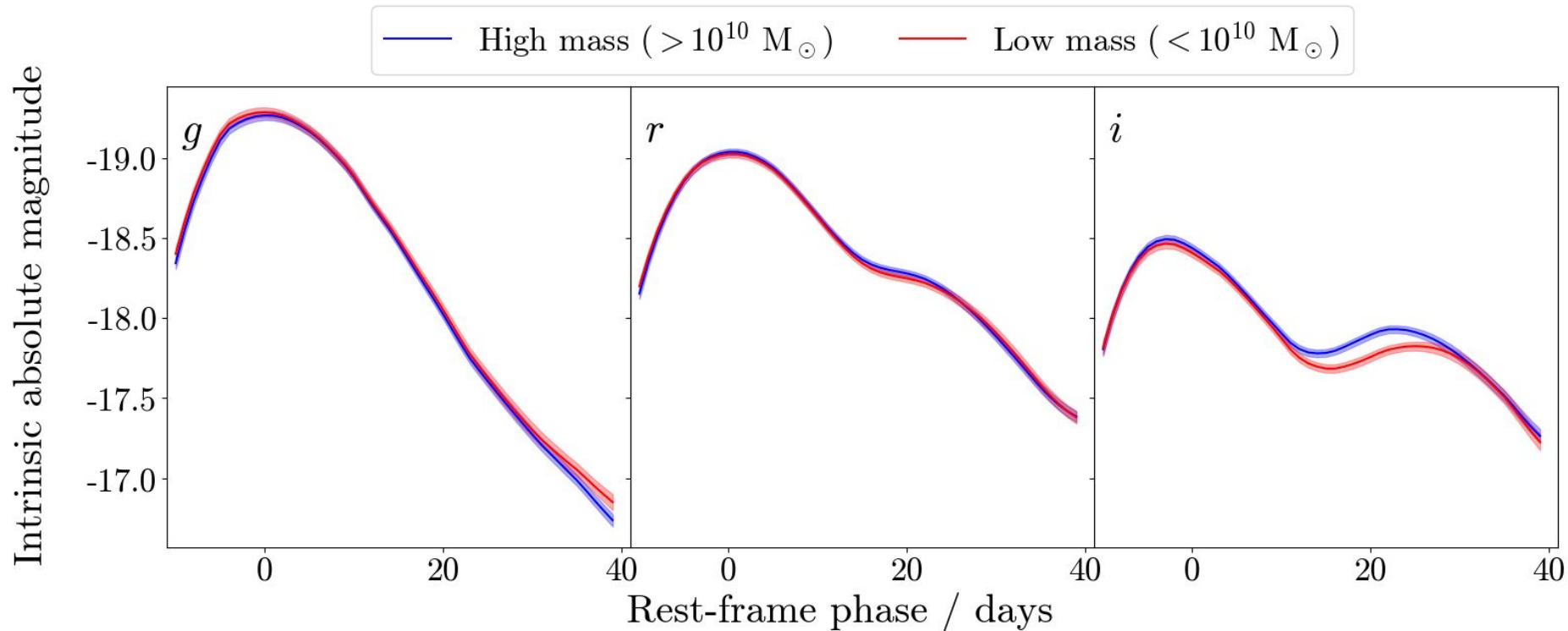


Grayling & Popovic (2024)

DES3YR + Foundation + PS1MD

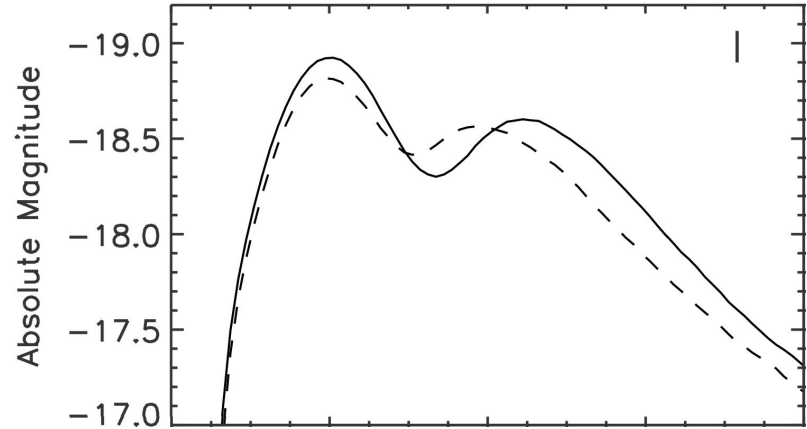


Grayling+2024



What's Causing the i-band Difference?

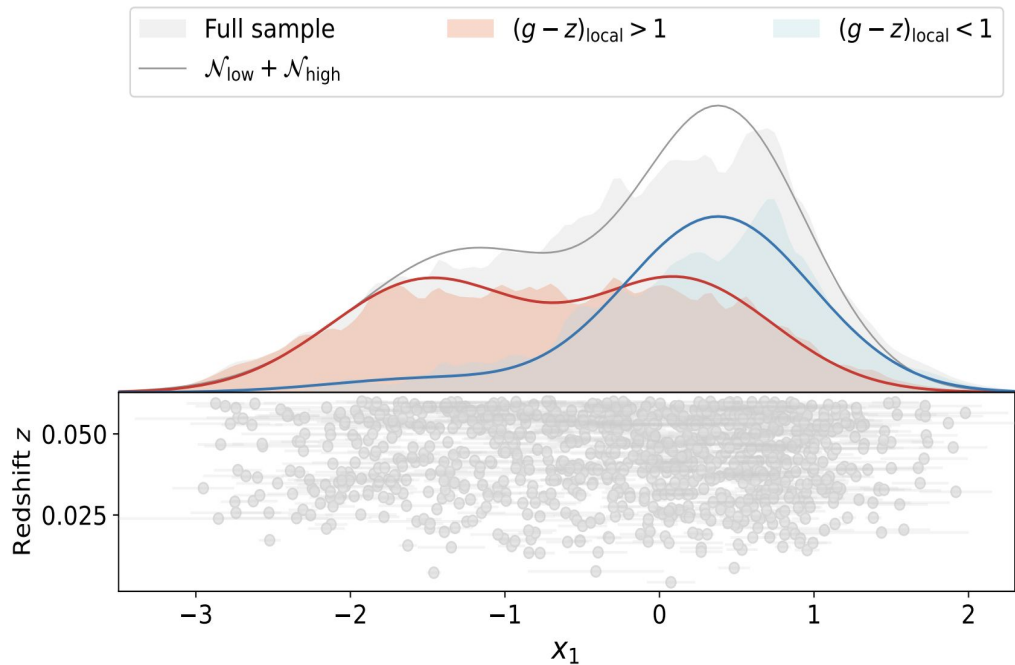
- *Something* is causing an intrinsic difference between SNe Ia in different environments, specifically in i-band.
- Kasen (2006) predicts that metallicity will have stronger impact on secondary maximum than around peak.
- Difference specifically in i-band; Deckers et al. (2024) find differences between behaviour of second maximum in i and r-band.



From Kasen (2006); dashed line = higher metallicity

Understanding the Environmental Dependence

- Mass can't be driving the difference, must be a proxy for something
- We can learn more from looking at the local environments in which SNe Ia explode
- Could be related to progenitor age or metallicity



Ginolin+2025

Could the SN Ia population evolve with redshift?

- The universe evolves with time
- Even if type Ia supernovae have constant properties, what if the dust in their host galaxies changes over time?
- We can approach this problem with BayeSN

$$R_V \sim N(\mu_R, \sigma_R^2)$$



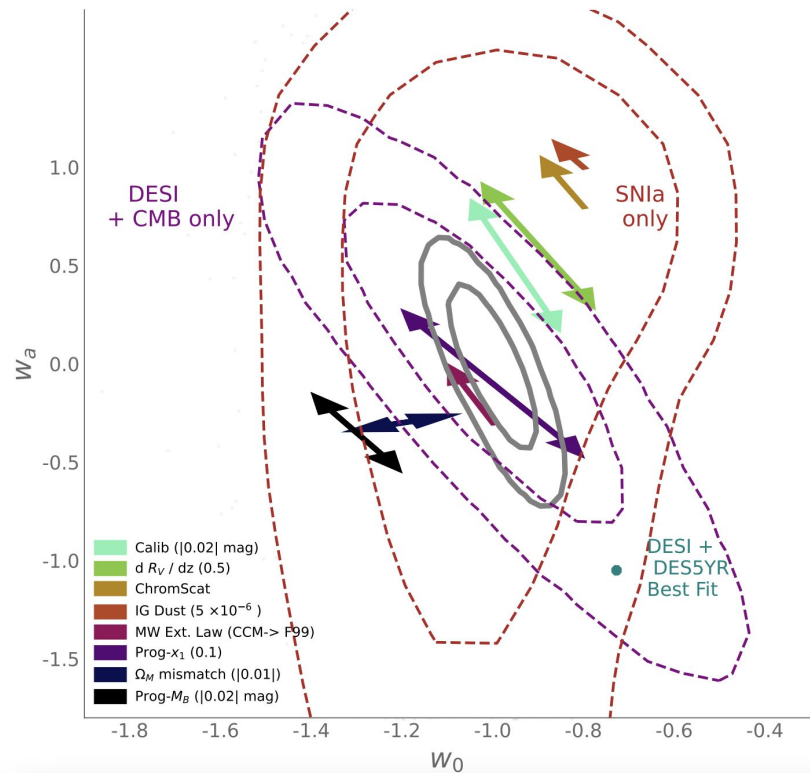
$$R_V \sim N(\mu_R(z), \sigma_R^2)$$

$$\mu_R(z) = \mu_{R,z0} + \eta_R z$$

It's really hard to tell

Parameter	Value
$\mu_{R,z0}$	2.58 ± 0.17
η_R	-0.38 ± 0.70

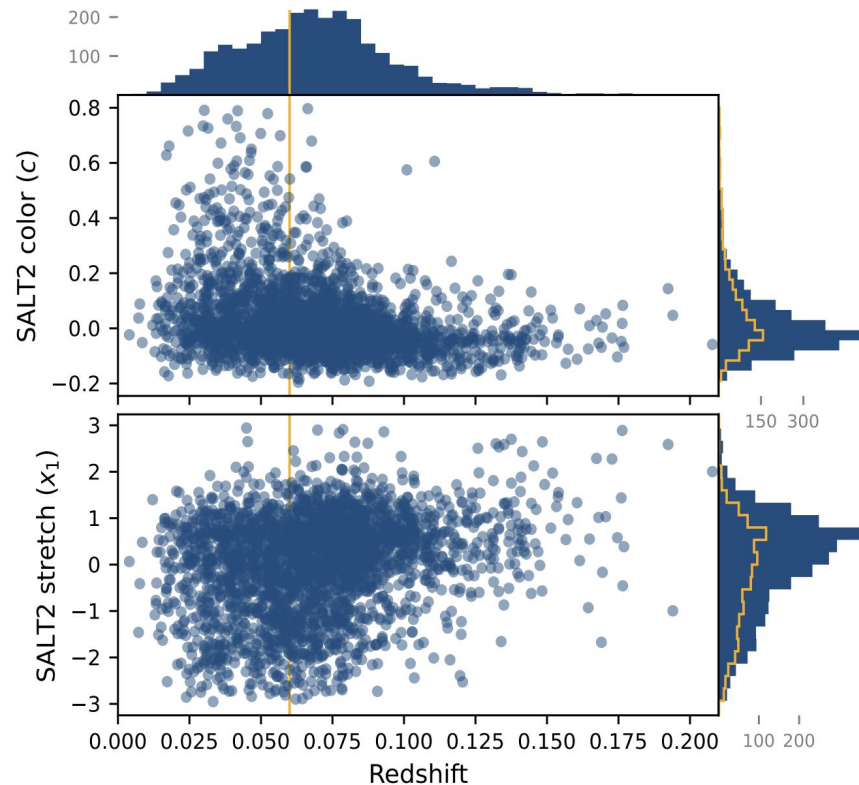
From Grayling+2024, based on 475 type Ia supernovae at $z < 0.4$



Dhawan+2024

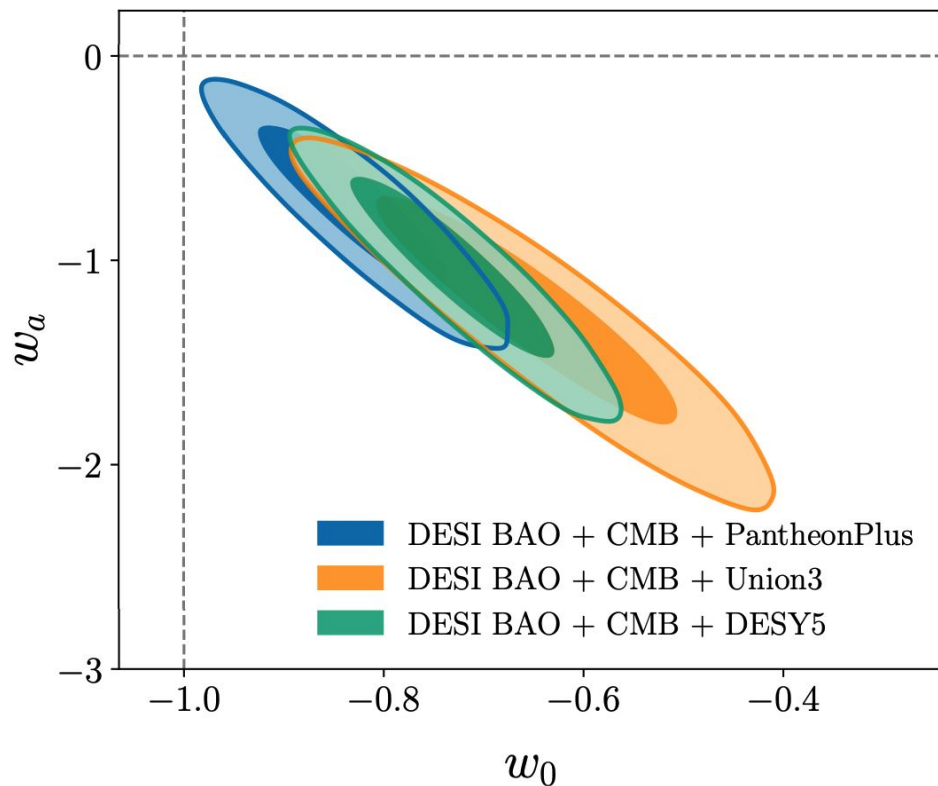
Prospects for probing redshift evolution

- Disentangling redshift evolution from selection effects is really challenging
- Need good volume limited samples
- An increasing number of samples with good samples minimally impacted by selection effects (e.g. ZTF) are now available



Implications for SN Cosmology

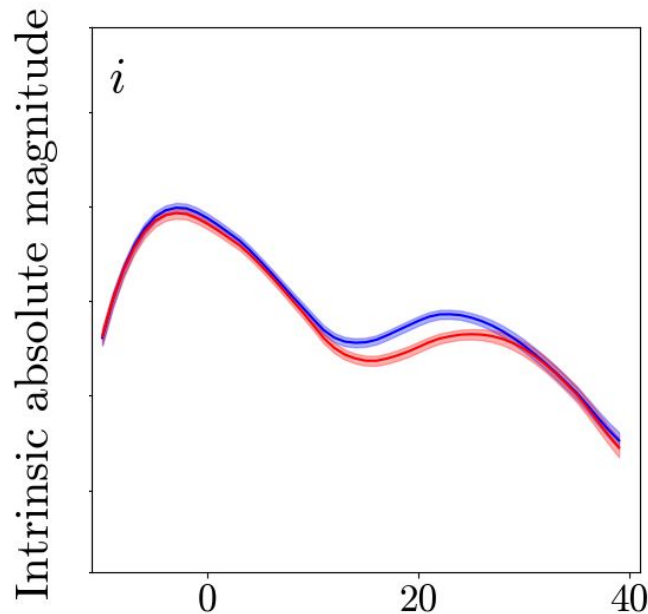
- Supernova cosmology is reliant on simulations - we need to get the forward model right
- We need to uncover exactly what is driving the environmental dependence that we observe
- Also vital to constrain any evolution of dust properties with redshift
- BayeSN can play a vital role in solving these problems



DESI Collaboration
2024

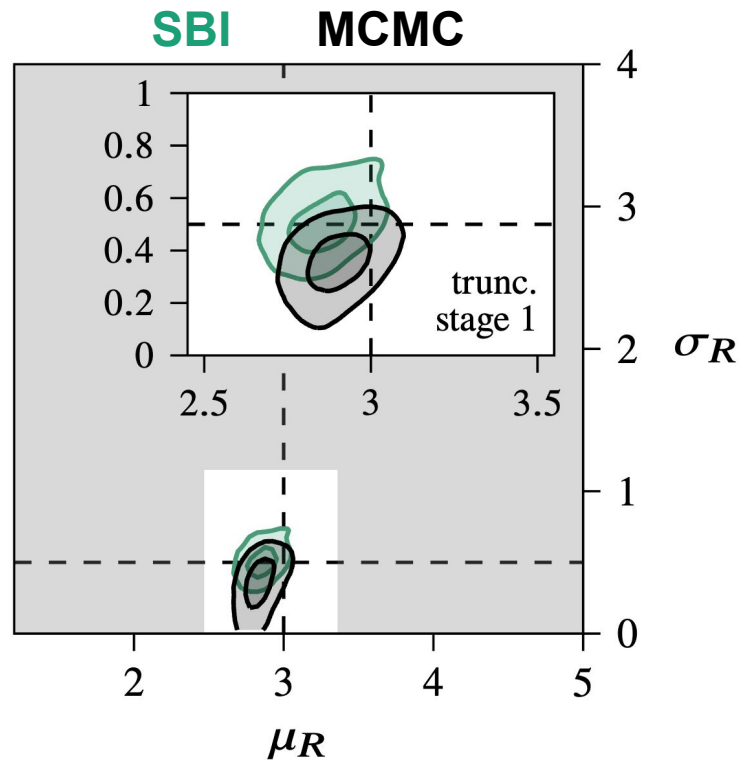
Plans for the Future

- Applying BayeSN to both spectra and photometry
 - Hopefully we can isolate differences in specific spectral features
 - Will provide more physical insight, better understanding of what is driving i-band maximum difference



Plans for the Future

- Cosmology with BayeSN
 - Currently working on plugging BayeSN into 'standard' SN cosmology pipeline
 - Developing new, principled methods for doing supernova cosmology using Simulation-based Inference (SBI)
 - SBI is the key to inferring cosmological parameters jointly with properties of SNe Ia



Karchev, Grayling et al. (2024)

Summary

- With BayeSN, we find evidence to support an intrinsic component of the mass step though dust may also play a role
- We find a significant environmental-dependence of the intrinsic properties of SNe Ia, specifically on the i-band secondary maximum
- Current data cannot yet constrain supernova host galaxy dust evolution with redshift
- Lots of ongoing work for BayeSN e.g. training on spectra, moving towards cosmology



MNRAS paper



New arxiv paper